

Jonathan Clayton

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Personal Statement

I am an expert in Natural Language Processing, currently completing a PhD at the University of Sheffield. My thesis, which is focused on generating graphical summaries of argumentative text, has been submitted, and I am due to sit my viva in September.

The technical core of my research has involved fine-tuning large language models using PyTorch. Alongside this, I designed a novel task within my subfield of NLP, and created and annotated corpora to support it.

I am the first author of three peer-reviewed papers published in leading NLP and speech processing venues, including *COMMA* (Computational Models of Argument), the 2023 Argument Mining workshop at *COLING*, and *Interspeech*.

Interests/ Specialisms

- Natural Language Processing
- Argument Mining & Computational Models of Argument
- Text Summarisation
- Large Language Models (fine-tuning, prompting, evaluation)
- Corpus Design & Annotation
- Speech Processing (ASR / TTS)

Technical Skills

- Python – PyTorch, Tensorflow, Scikit-Learn, Pandas, PySpark
- HuggingFace / OpenAI API
- Git, GitLab
- High Performance Computing / GPU clusters
- Linux / BASH scripting
- SQL
- C++, Java, R

Education

University of Sheffield 2020 – Present

PhD in the Summarisation of Natural Language Argument

- Trained LLMs for Argument Mining tasks using PyTorch
- Proposed a novel task within the field of Argument Mining (Argument Summary Graph Parsing)
- Developed three novel data resources for Argument Mining

University of Edinburgh 2018-2019

MSc in Speech and Language Processing (Passed with Distinction)

- Thesis on decoding speech from brainwaves using a wearable EEG device
- Coursework on a variety of topics in Machine Learning, including Natural Language Processing, Automatic Speech Recognition and Speech Synthesis

University of York 2012 – 2015

BA in Linguistics (Passed with First Class Degree)

- Thesis researching speech perception in learners of Spanish

Internships

Internship with Digital Humanities Institute [🔗](#) *University of Sheffield*

- I devised a novel implementation of a process for extracting “quads” (a type of linguistic unit) from a huge historical text corpus using GPUs and PyTorch.
- Proposed methods for analysing the resulting data using LLMs and clustering

June 2024 – Sep 2024

algorithms.

Internship with [SpeakUnique](#)

- Produced a state of the art dataset for the task of silent speech detection (detecting speech from brainwaves using EEG devices)
- Carried out research into Machine Learning techniques for carrying out this task

Edinburgh, UK
Oct 2019 - Jan 2020

Publications

Parsing Graphical Summaries from Argumentative Dialogues

October 2024

Jonathan Clayton, Marco Damonte, Robert Gaizauskas
Computational Models of Argument
DOI: [10.3233/FAIA240308](https://doi.org/10.3233/FAIA240308) 

Predicting the Presence of Reasoning Markers in Argumentative Text

October 2022

Jonathan Clayton, Robert Gaizauskas
Proceedings of the 9th Workshop on Argument Mining
Link: <https://aclanthology.org/2022.argmining-1.13/> 

Decoding Imagined, Heard, and Spoken Speech: Classification and Regression of EEG Using a 14-Channel Dry-Contact Mobile Headset

October 2020

Jonathan Clayton, Scott Wellington, Cassia Valentini-Botinhao, Oliver Watts
Interspeech 2020
DOI: [10.21437/Interspeech.2020-2745](https://doi.org/10.21437/Interspeech.2020-2745) 

Work Experience

Graduate Teaching Assistant *University of Sheffield*

2021 - 2023

Helped out in labs and marked exam papers for several modules, including Robotics, Artificial Intelligence and Natural Language Processing.

English Teacher

2017-2019

Language Link, Yihai Primary School, Beijing, China

Taught English, as well as integrated mathematics and art classes, to primary school students.

English Teacher

2015-2017

Akerle Academy, Ermua, Bizkaia, Spain

Taught English in a tutoring centre to Spanish students of all ages.

Laboratory Assistant

Summer 2012, 2013, 2014

SATRA Technology Centre, Kettering, United Kingdom

Assisted technicians in carrying out product safety tests

Other

Languages (human): Spanish (C1 Level), Mandarin Chinese (B1)

Free time: I enjoy long distance hiking. Routes I have walked include the [Pennine Way](#)  and the English [Coast to Coast Walk](#) .